

AI Model Zoo Module For SPIDHIVE

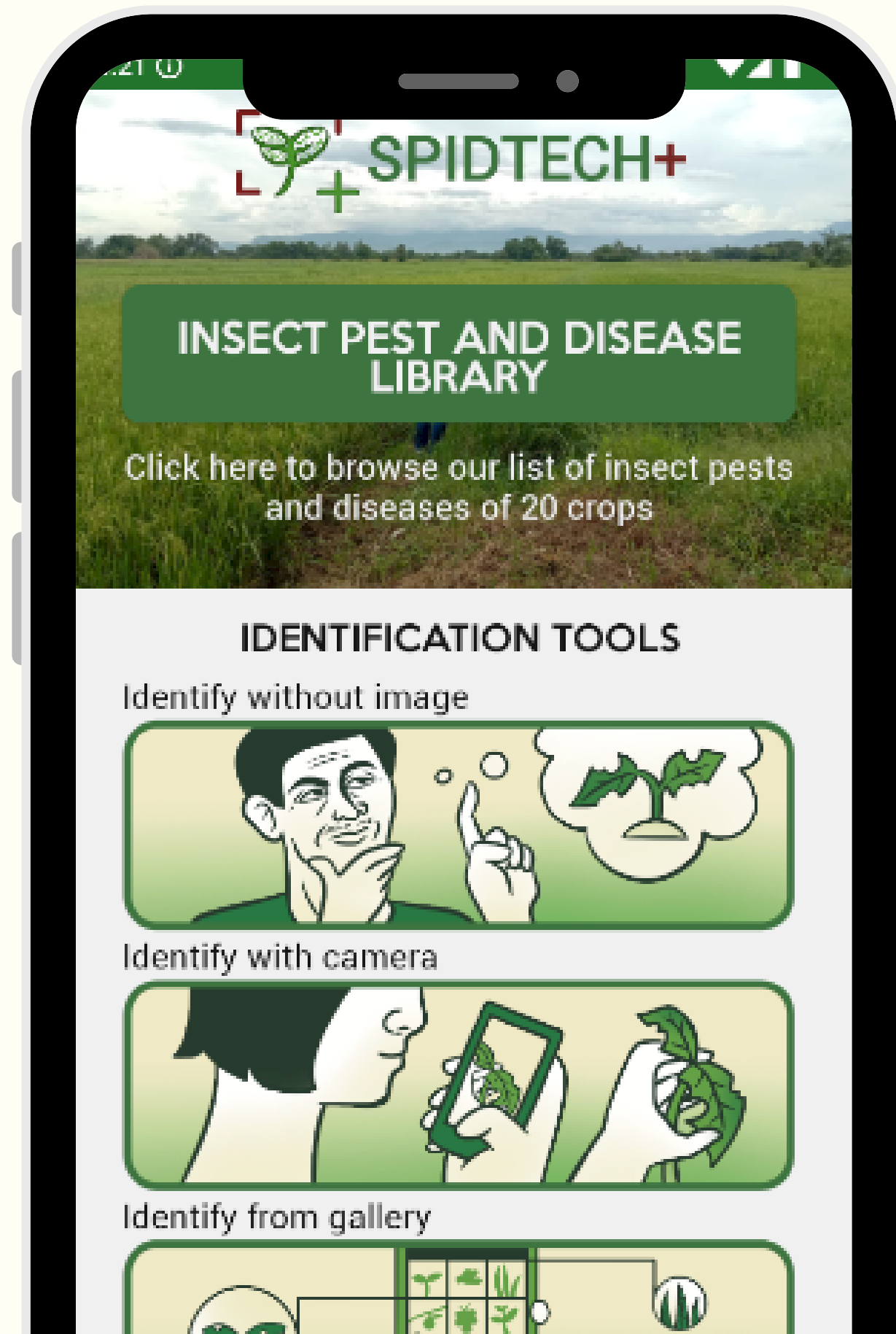


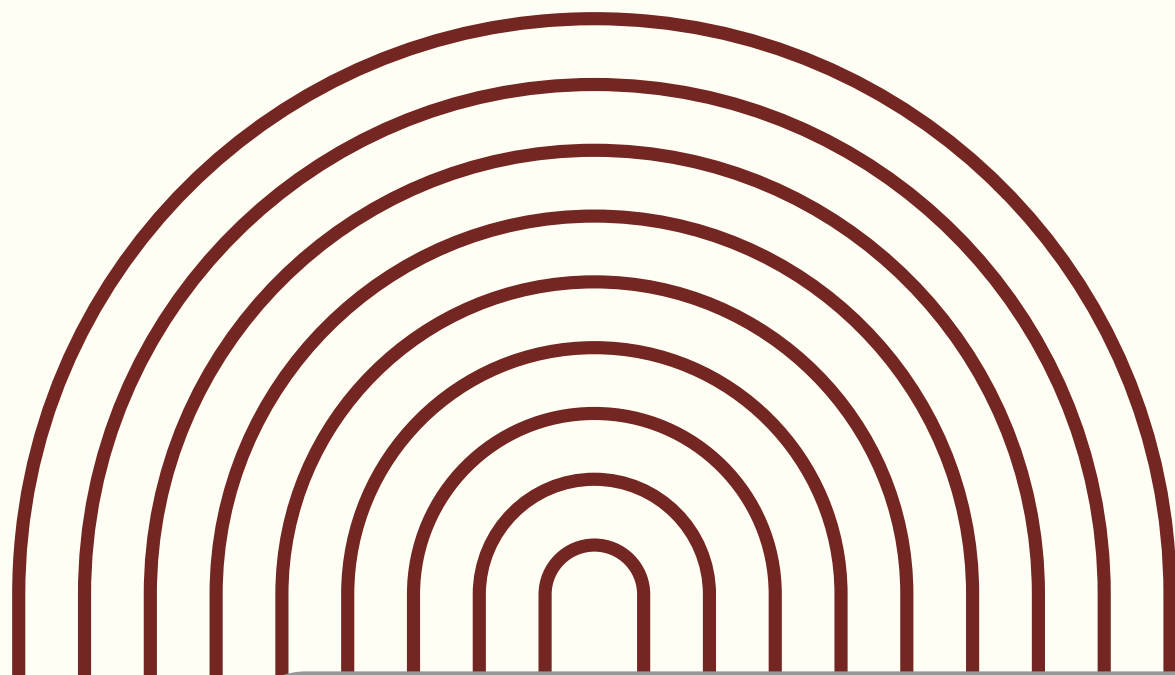
NADINE ARABEL ANARETA & VAL RANDOLF MADRID



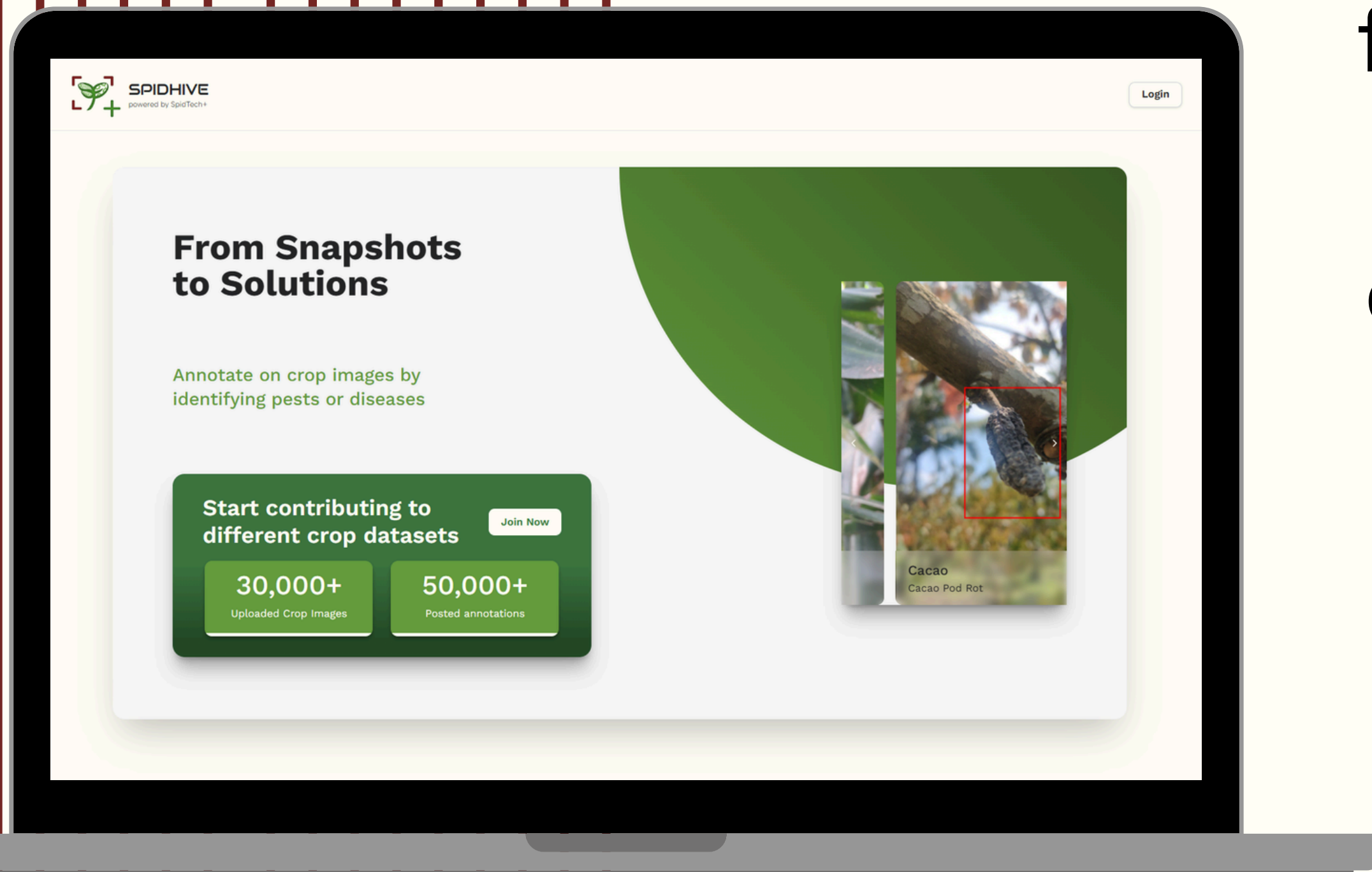
Staple crops are highly vulnerable to pests and diseases, which severely disrupt our food supply

SPIDTECH+ allows farmers to remotely monitor and identify these threats using AI.





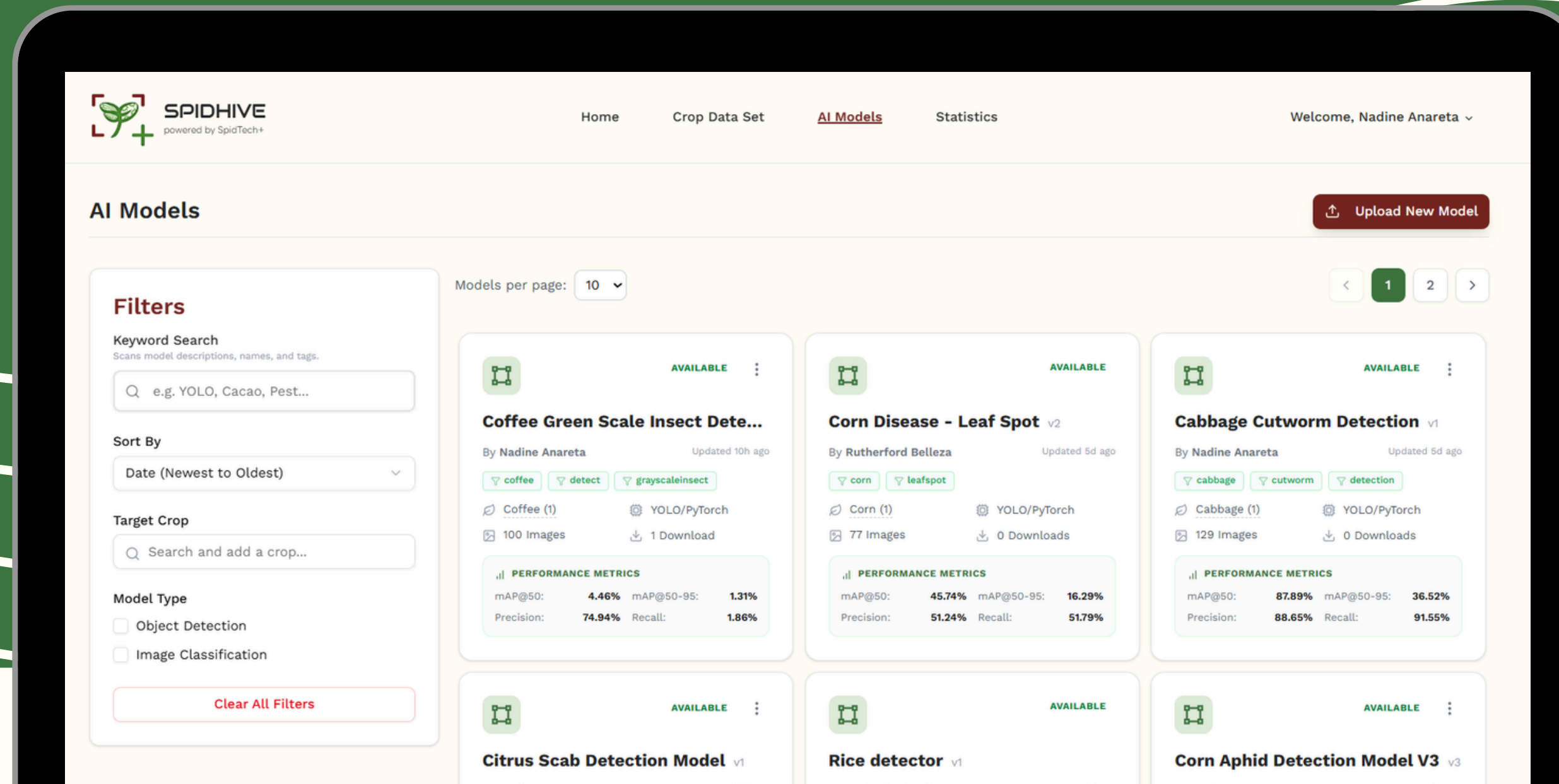
Data collected
from the mobile
app is sent to a
centralized web
platform called
SPIDHIVE.



Lacked a
**centralized
mechanism** to
manage, store, and
deploy the actual AI
models themselves

**No streamlined
pipeline** to get an
updated AI model
from SPIDHIVE to
SPIDTECH+

Localized AI Model Zoo integrated directly into SPIDHIVE



The screenshot displays the SPIDHIVE AI Models interface. The header includes the SPIDHIVE logo (powered by SpidTech+), navigation links (Home, Crop Data Set, AI Models, Statistics), and a user greeting (Welcome, Nadine Anareta). The main section is titled "AI Models" and features a "Filters" sidebar on the left. The sidebar includes a "Keyword Search" field, a "Sort By" dropdown (set to "Date (Newest to Oldest)"), a "Target Crop" search field, and "Model Type" checkboxes for "Object Detection" and "Image Classification". A "Clear All Filters" button is at the bottom of the sidebar. The main content area shows a grid of model cards, each with a crop icon, a title, author, update date, tags, target crop, image count, download count, and performance metrics (mAP@50, mAP@50-95, Precision, Recall). The cards are for "Coffee Green Scale Insect Dete...", "Corn Disease - Leaf Spot v2", "Cabbage Cutworm Detection v1", "Citrus Scab Detection Model v1", "Rice detector v1", and "Corn Aphid Detection Model V3 v3".

Filters

Keyword Search
Scans model descriptions, names, and tags.

Search: e.g. YOLO, Cacao, Pest...

Sort By
Date (Newest to Oldest)

Target Crop
Search and add a crop...

Model Type
☐ Object Detection
☐ Image Classification

Clear All Filters

Models per page: 10

Coffee Green Scale Insect Dete...
By Nadine Anareta Updated 10h ago
Tags: coffee, detect, grayscaleinsect
Target Crop: Coffee (1)
Images: 100 Images Downloads: 1 Download
YOLO/PyTorch
PERFORMANCE METRICS
mAP@50: 4.46% mAP@50-95: 1.31%
Precision: 74.94% Recall: 1.86%

Corn Disease - Leaf Spot v2
By Rutherford Belleza Updated 5d ago
Tags: corn, leafspot
Target Crop: Corn (1)
Images: 77 Images Downloads: 0 Downloads
YOLO/PyTorch
PERFORMANCE METRICS
mAP@50: 45.74% mAP@50-95: 16.29%
Precision: 51.24% Recall: 51.79%

Cabbage Cutworm Detection v1
By Nadine Anareta Updated 5d ago
Tags: cabbage, cutworm, detection
Target Crop: Cabbage (1)
Images: 129 Images Downloads: 0 Downloads
YOLO/PyTorch
PERFORMANCE METRICS
mAP@50: 87.89% mAP@50-95: 36.52%
Precision: 88.65% Recall: 91.55%

Citrus Scab Detection Model v1

Rice detector v1

Corn Aphid Detection Model V3 v3

2 ZERO
HUNGER



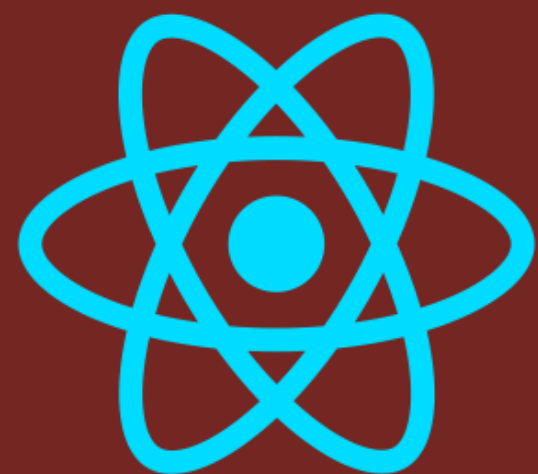
15 LIFE
ON LAND

Controlling
transboundary pests
and diseases is
essential to double
agricultural
productivity and
ensure safe

15 LIFE
ON LAND



Invasive pests and
pathogens threaten
terrestrial
ecosystems



React JS



SPIDHIVE

Stores and Manages AI Models

downloads
datasets and
used for training

Available Datasets

validated by
experts

Displays images for
expert evaluation

SUPABASE

Stores AI Models from
SPIDHIVE and Syncs with
SPIDTECH+

Stores the images from
SPIDTECH+ and Syncs
with SPIDHIVE

SPIDTECH+

Fetches AI Models

Uses the models and
generates feedback or
report images



From Snapshots to Solutions

Annotate on crop images by
identifying pests or diseases

**Start contributing to
different crop datasets**

[Join Now](#)

30,000+

Uploaded Crop Images

50,000+

Posted annotations



Rice

Rice False Smut

AI Models

[Upload New Model](#)

Filters

Keyword Search

Scans model descriptions, names, and tags.

RECENT

yolo

Sort By

Date (Newest to Oldest) ▼

Target Crop

Model Type

- ☐ Object Detection
- ☐ Image Classification

[Clear All Filters](#)

Models per page: 10 ▼

< 1 2 >



AVAILABLE



Coffee Green Scale Insect Dete...

By Nadine Anareta

Updated 5d ago

coffee detect grayscaleinsect

Coffee (1)

YOLO/PyTorch

100 Images

1 Download

PERFORMANCE METRICS

mAP@50:	4.46%	mAP@50-95:	1.31%
Precision:	74.94%	Recall:	1.88%



AVAILABLE



Corn Disease - Leaf Spot v2

By Rutherford Belleza

Updated 10d ago

corn leafspot

Corn (1)

YOLO/PyTorch

77 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	45.74%	mAP@50-95:	16.29%
Precision:	51.24%	Recall:	51.79%



AVAILABLE



Cabbage Cutworm Detection v1

By Nadine Anareta

Updated 10d ago

cabbage cutworm detection

Cabbage (1)

YOLO/PyTorch

129 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	87.89%	mAP@50-95:	36.52%
Precision:	88.65%	Recall:	91.55%



AVAILABLE



Citrus Scab Detection Model v1

By Nadine Anareta

Updated 12d ago

orange lime sour

Citrus (1)

YOLO/PyTorch

146 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	36.28%	mAP@50-95:	15.12%
Precision:	48.6%	Recall:	42.42%



AVAILABLE



Rice detector v1

By Rutherford Belleza

Updated 10d ago

rice model

Rice (1)

YOLO/PyTorch

120 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	25.06%	mAP@50-95:	5.41%
Precision:	58.57%	Recall:	20.62%



AVAILABLE



Corn Aphid Detection Model V3 v3

By Nadine Anareta

Updated 5d ago

corn pest aphid detection

Corn (1)

YOLO11m

97 Images

2 Downloads

PERFORMANCE METRICS

mAP@50:	20.27%	mAP@50-95:	5.22%
Precision:	36.43%	Recall:	29.33%

AI Models

[Upload New Model](#)

Filters

Keyword Search

Scans model descriptions, names, and tags.

RECENT

yolo

Sort By

Date (Newest to Oldest)

Target Crop

Model Type

- ☐ Object Detection
- ☐ Image Classification

[Clear All Filters](#)

Models per page: 10

< 1 2 >



AVAILABLE



Coffee Green Scale Insect Dete...

By Nadine Anareta

Updated 5d ago

coffee

detect

grayscaleinsect

Coffee (1)

YOLO/PyTorch

100 Images

1 Download

PERFORMANCE METRICS

mAP@50: 4.46% mAP@50-95: 1.31%

Precision: 74.94% Recall: 1.86%



AVAILABLE



Corn Disease - Leaf Spot v2

By Rutherford Belleza

Updated 10d ago

corn

leafspot

Corn (1)

YOLO/PyTorch

77 Images

0 Downloads

PERFORMANCE METRICS

mAP@50: 45.74% mAP@50-95: 16.29%

Precision: 51.24% Recall: 51.79%



AVAILABLE



Cabbage Cutworm Detection v1

By Nadine Anareta

Updated 10d ago

cabbage

cutworm

detection

Cabbage (1)

YOLO/PyTorch

129 Images

0 Downloads

PERFORMANCE METRICS

mAP@50: 87.89% mAP@50-95: 36.52%

Precision: 88.65% Recall: 91.55%



AVAILABLE



Citrus Scab Detection Model v1

By Nadine Anareta

Updated 12d ago

orange

lime

sour

Citrus (1)

YOLO/PyTorch

146 Images

0 Downloads

PERFORMANCE METRICS

mAP@50: 36.28% mAP@50-95: 15.12%

Precision: 48.6% Recall: 42.42%



AVAILABLE



Rice detector v1

By Rutherford Belleza

Updated 10d ago

rice

model

Rice (1)

YOLO/PyTorch

120 Images

0 Downloads

PERFORMANCE METRICS

mAP@50: 25.06% mAP@50-95: 5.41%

Precision: 58.57% Recall: 20.62%



AVAILABLE



Corn Aphid Detection Model V3...

By Nadine Anareta

Updated 5d ago

corn

pest

aphid

detection

Corn (1)

YOLO11m

97 Images

2 Downloads

PERFORMANCE METRICS

mAP@50: 20.27% mAP@50-95: 5.22%

Precision: 36.43% Recall: 29.33%

AI Models

[Upload New Model](#)

Filters

Keyword Search

Scans model descriptions, names, and tags.

RECENT

yolo

Sort By

Date (Newest to Oldest) ▼

Target Crop

Model Type

- ☐ Object Detection
- ☐ Image Classification

[Clear All Filters](#)

Models per page: 10 ▼

< 1 2 >



AVAILABLE



Corn Aphid Detection M v4

By Nadine Anareta

just now

▼ corn ▼ aphid ▼ detection



Corn (1)

YOLO11M

97 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	20.27%	mAP@50-95:	5.22%
Precision:	36.43%	Recall:	29.33%



AVAILABLE



Coffee Green Scale Insect Det...

By Nadine Anareta

Updated 5d ago

Classes:

CoffeeGreenScaleInsect

▼ grayscaleinsect

Coffee (1)

YOLO/PyTorch

100 Images

1 Download

PERFORMANCE METRICS

mAP@50:	4.46%	mAP@50-95:	1.31%
Precision:	74.94%	Recall:	1.86%



AVAILABLE

Corn Disease - Leaf Spot v2

By Rutherford Belleza

Updated 10d ago

▼ corn ▼ leafspot

Corn (1)

YOLO/PyTorch

77 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	45.74%	mAP@50-95:	16.29%
Precision:	51.24%	Recall:	51.79%



AVAILABLE



Cabbage Cutworm Detection v1

By Nadine Anareta

Updated 10d ago

▼ cabbage ▼ cutworm ▼ detection

Cabbage (1)

YOLO/PyTorch

129 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	87.89%	mAP@50-95:	36.52%
Precision:	88.65%	Recall:	91.55%



AVAILABLE



Citrus Scab Detection Model v1

By Nadine Anareta

Updated 12d ago

▼ orange ▼ lime ▼ sour

Citrus (1)

YOLO/PyTorch

146 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	36.28%	mAP@50-95:	15.12%
Precision:	48.6%	Recall:	42.42%



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Rice detector v1

By Rutherford Belleza

Updated 10d ago

▼ rice ▼ model

Rice (1)

YOLO/PyTorch

120 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	25.06%	mAP@50-95:	5.41%
Precision:	58.57%	Recall:	20.62%

AI Models

[Upload New Model](#)

Filters

Keyword Search

Scans model descriptions, names, and tags.

RECENT

[detect](#) [grayscaleinsect](#) [rice](#) [cabbage](#) [orange](#)

Sort By

[Date \(Newest to Oldest\)](#)

Target Crop

Model Type

- ☐ Object Detection
- ☐ Image Classification

[Clear All Filters](#)Models per page: 10 ▼[<](#) [1](#) [2](#) [>](#)

AVAILABLE



Corn Aphid Detection M v4

By Nadine Anareta

16h ago

[corn](#) [aphid](#) [detection](#)

Corn (1) YOLO11M

97 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	20.27%	mAP@50-95:	5.22%
Precision:	36.43%	Recall:	29.33%



AVAILABLE



Coffee Green Scale Insect Dete...

By Nadine Anareta

updated 5d ago

[coffee](#) [detect](#) [grayscaleinsect](#)

Coffee (1) YOLO/PyTorch

100 Images 1 Download

PERFORMANCE METRICS

mAP@50:	4.46%	mAP@50-95:	1.31%
Precision:	74.94%	Recall:	1.86%



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Corn Disease - Leaf Spot v2

By Rutherford Belleza

updated 11d ago

[corn](#) [leafspot](#)

Corn (1) YOLO/PyTorch

77 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	45.74%	mAP@50-95:	16.29%
Precision:	51.24%	Recall:	51.79%



AVAILABLE



Cabbage Cutworm Detection v1

By Nadine Anareta

Updated 11d ago

[cabbage](#) [cutworm](#) [detection](#)

Cabbage (1) YOLO/PyTorch

128 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	87.89%	mAP@50-95:	36.52%
Precision:	88.65%	Recall:	91.55%



AVAILABLE



Citrus Scab Detection Model v1

By Nadine Anareta

Updated 12d ago

[orange](#) [lime](#) [sour](#)

Citrus (1) YOLO/PyTorch

146 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	36.28%	mAP@50-95:	15.12%
Precision:	48.6%	Recall:	42.42%



AVAILABLE

Rice detector v1

By Rutherford Belleza

Updated 11d ago

[rice](#) [model](#)

Rice (1) YOLO/PyTorch

120 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	25.06%	mAP@50-95:	5.41%
Precision:	58.57%	Recall:	20.62%

[← Back to Models](#)[✎ Edit Details](#)[🗑 Delete](#)[📄 Download Source ZIP](#)[📊 Overview](#)[🖼 Images](#)

97

[📁 Versions](#)[📈 Analytics](#)

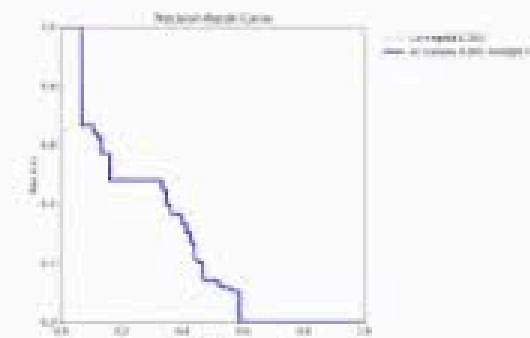
Analytics

PRIMARY METRIC (MAP@50)

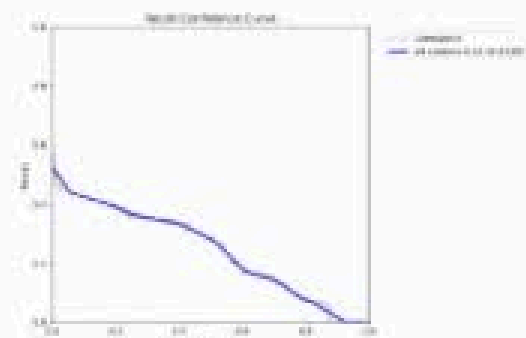
20.3% 97 Images  0 Downloads

MAP@50	20.3%
MAP@50-95	5.2%
PRECISION	36.4%
RECALL	29.3%

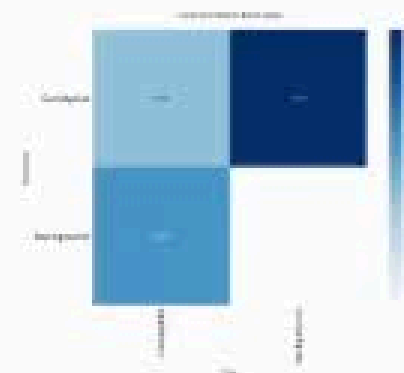
Evaluation Visuals 6 Graphs



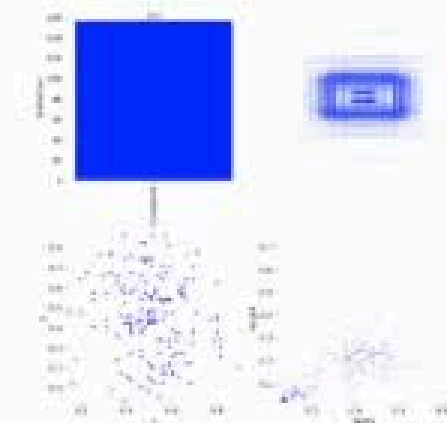
BOXPR CURVE



BOXR CURVE



CONFUSION MATRIX NORMALIZED



LABELS

AI Models

[Upload New Model](#)

Filters

Keyword Search

Scans model descriptions, names, and tags.

RECENT

[detect](#) [grayscaleinsect](#) [rice](#) [cabbage](#) [orange](#)

Sort By

Target Crop

Model Type

- ☐ Object Detection
- ☐ Image Classification

[Clear All Filters](#)Models per page: 10 ▼< 1 2 >

AVAILABLE



Coffee Green Scale Insect Dete...

By Nadine Anareta

Updated 5d ago

[coffee](#) [detect](#) [grayscaleinsect](#)

Coffee (1)

YOLO/PyTorch

100 Images

1 Download

PERFORMANCE METRICS

mAP@50:	4.46%	mAP@50-95:	1.31%
Precision:	74.94%	Recall:	1.88%



AVAILABLE



Corn Disease - Leaf Spot v2

By Rutherford Belleza

Updated 11d ago

[corn](#) [leafspot](#)

Corn (1)

YOLO/PyTorch

77 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	45.74%	mAP@50-95:	16.29%
Precision:	51.24%	Recall:	51.79%



AVAILABLE



Cabbage Cutworm Detection v1

By Nadine Anareta

Updated 11d ago

[cabbage](#) [cutworm](#) [detection](#)

Cabbage (1)

YOLO/PyTorch

129 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	87.89%	mAP@50-95:	36.52%
Precision:	88.65%	Recall:	91.55%



AVAILABLE



Citrus Scab Detection Model v1

By Nadine Anareta

Updated 12d ago

[orange](#) [lime](#) [sour](#)

Citrus (1)

YOLO/PyTorch

146 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	36.28%	mAP@50-95:	15.12%
Precision:	48.6%	Recall:	42.42%



AVAILABLE



Rice detector v1

By Rutherford Belleza

Updated 11d ago

[rice](#) [model](#)

Rice (1)

YOLO/PyTorch

120 Images

0 Downloads

PERFORMANCE METRICS

mAP@50:	25.06%	mAP@50-95:	5.41%
Precision:	58.57%	Recall:	20.62%



AVAILABLE



Corn Aphid Detection Model V3 v3

By Nadine Anareta

Updated 5d ago

[corn](#) [pest](#) [aphid](#) [detection](#)

Corn (1)

YOLO11m

97 Images

2 Downloads

PERFORMANCE METRICS

mAP@50:	20.27%	mAP@50-95:	5.22%
Precision:	36.43%	Recall:	29.33%

AI Models

[Upload New Model](#)

Filters

Keyword Search

Scans model descriptions, names, and tags.

RECENT

[detect](#) [grayscaleinsect](#) [rice](#) [cabbage](#) [orange](#)

Sort By

Target Crop

Model Type

- ☐ Object Detection
- ☐ Image Classification

[Clear All Filters](#)Models per page: 10 ▼< 1 2 >

AVAILABLE



Coffee Green Scale Insect Dete...

By Nadine Anareta

Updated 5d ago

[coffee](#) [detect](#) [grayscaleinsect](#)

Coffee (1) YOLO/PyTorch

100 Images 1 Download

PERFORMANCE METRICS

mAP@50:	4.46%	mAP@50-95:	1.31%
Precision:	74.94%	Recall:	1.88%



AVAILABLE



Corn Disease - Leaf Spot v2

By Rutherford Belleza

Updated 11d ago

[corn](#) [leafspot](#)

Corn (1) YOLO/PyTorch

77 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	45.74%	mAP@50-95:	16.29%
Precision:	51.24%	Recall:	51.79%



AVAILABLE



Cabbage Cutworm Detection v1

By Nadine Anareta

Updated 11d ago

[cabbage](#) [cutworm](#) [detection](#)

Cabbage (1) YOLO/PyTorch

129 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	87.89%	mAP@50-95:	36.52%
Precision:	88.65%	Recall:	91.55%



AVAILABLE



Citrus Scab Detection Model v1

By Nadine Anareta

Updated 12d ago

[orange](#) [lime](#) [sour](#)

Citrus (1) YOLO/PyTorch

146 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	36.28%	mAP@50-95:	15.12%
Precision:	48.6%	Recall:	42.42%



AVAILABLE



Rice detector v1

By Rutherford Belleza

Updated 11d ago

[rice](#) [model](#)

Rice (1) YOLO/PyTorch

120 Images 0 Downloads

PERFORMANCE METRICS

mAP@50:	25.06%	mAP@50-95:	5.41%
Precision:	58.57%	Recall:	20.62%



AVAILABLE



Corn Aphid Detection Model V3 v3

By Nadine Anareta

Updated 5d ago

[corn](#) [pest](#) [aphid](#) [detection](#)

Corn (1) YOLO11m

97 Images 2 Downloads

PERFORMANCE METRICS

mAP@50:	20.27%	mAP@50-95:	5.22%
Precision:	36.43%	Recall:	29.33%

Crop Data Set

Q Search

 Upload Crop Image

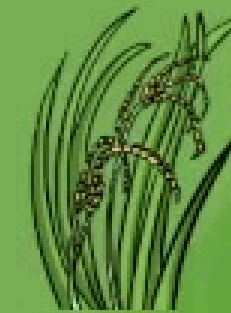


SPIDTECH+

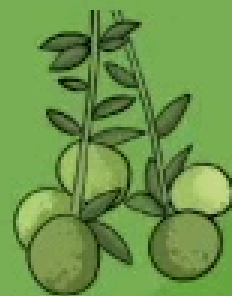
SPIDTECH+ Feed
5 Photos



Corn
5861 Photos



Rice
5813 Photos



Citrus
3372 Photos



Coffee
2754 Photos



Eggplant
2694 Photos

SUS

88

Excellent

Workflow Automation

Information Architecture & User Interface

System Security and Data Privacy

Retrieval and Navigation

UI Density and Context

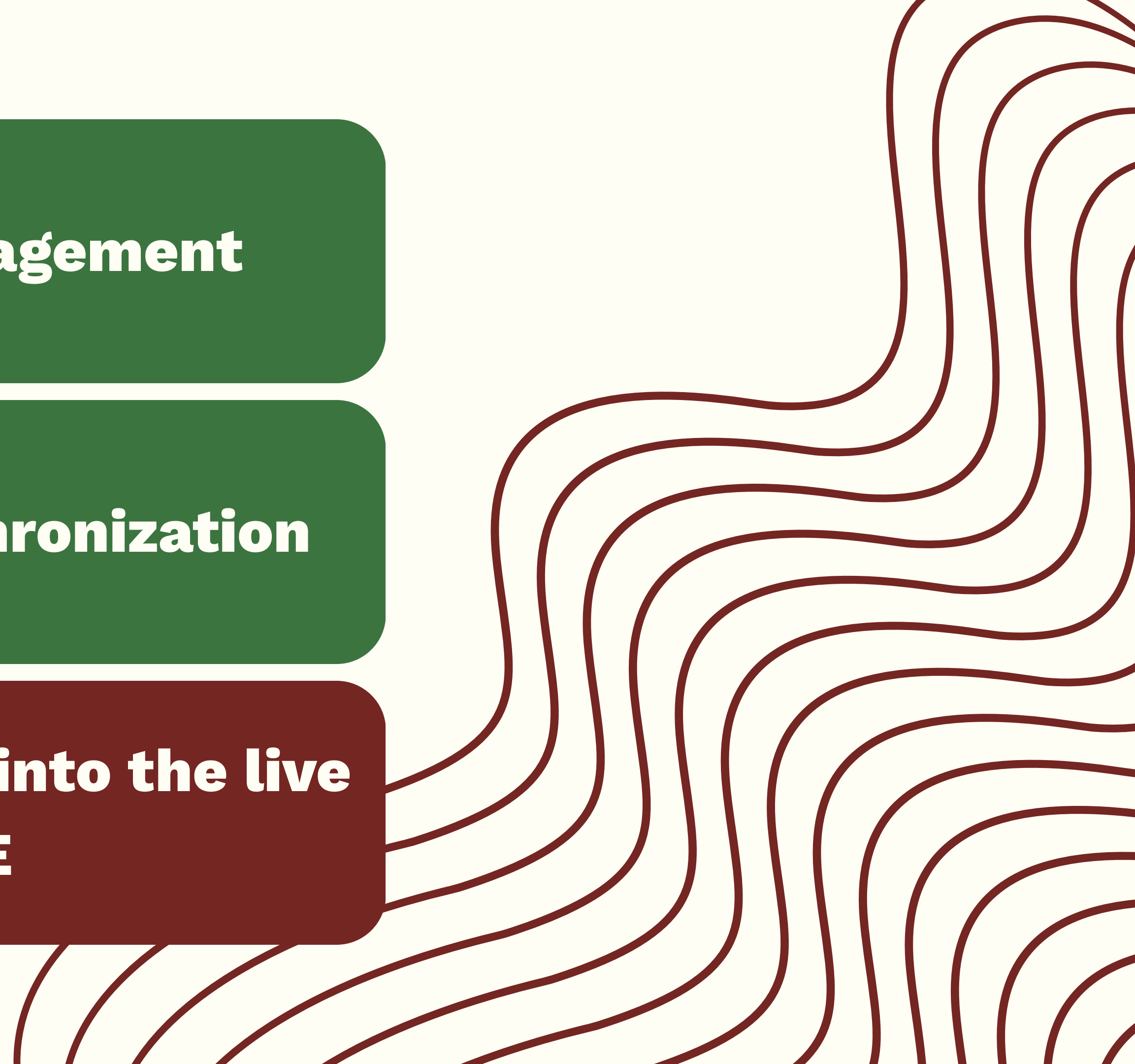
Search Filtering

Granular Inspection

Centralized Management

Bidirectional Synchronization

**Official deployment into the live
SPIDHIVE**

A series of dark red, wavy, horizontal lines that flow from the right side of the text boxes across the right half of the image, creating a sense of movement and connectivity.

AI Model Zoo Module For SPIDHIVE



NADINE ARABEL ANARETA & VAL RANDOLF MADRID